

$$\textcircled{1} \quad \text{True Altitude} = \text{Indicated Alt} + \frac{4 \times \text{temp Dev} \times \text{I.A}}{1000}$$

$$\textcircled{2} \quad \text{Temp Deviation} = \text{OAT} - \text{ISA}$$

$$\textcircled{3} \quad \text{Pressure Altitude} = \text{Elevation} + 30 (1013 - \text{QNH})$$

$$\textcircled{4} \quad \text{Pressure Altitude} = 30 \times (1013 - \text{QFE})$$

$$\textcircled{5} \quad \text{Density Altitude} = \text{PA} + 120 (\text{ISA Deviation})$$

$$\textcircled{6} \quad \text{Mach Number} = \frac{\text{TAS}}{\text{LSS}}$$

$$\textcircled{7} \quad \text{LSS} = 39 \sqrt{273 + T^{\circ}\text{C}}$$

$$\textcircled{8} \quad \text{TAT} = \text{SAT} (1 + 0.2 \text{KM}^2)$$

$$\textcircled{9} \quad \text{Transport Slander} = \frac{\text{Ground Speed} \times \tan \theta}{60}$$

$$\textcircled{10} \quad \text{Angle of turn} = 10\% \text{ of TAS} + 7 (\text{for rate 1})$$

$$\textcircled{11} \quad \text{Radius of turn} = \frac{V^2}{\tan \theta}$$

$V = \text{TAS in m/sec}$
 $g = 10 \text{ m/sec}^2$
 $\theta = \text{AOB}$

$$\textcircled{12} \quad \text{Rate of turn} = \frac{g \tan \theta}{V}$$

$$\textcircled{13} \quad \text{Speed of EMV}$$

186000 ST MILES / SEC

162000 NMS / SEC

3000000000 m / SEC

3000000 km / SEC

300 MILLION METRE / SEC

(13) $c = f \times \lambda$

(14) Relative Bearing + Magnetic Heading = QDM

(15) $\theta = \frac{S}{R} \times 60$ OR $ANGLE = \frac{\Delta OT}{\Delta h} \times 60$

(16) Time to station = $\frac{\text{Time change}}{\text{Bearing change}} \times 60^\circ$ upto 20°

(17) $HT = A \times R \times 101$

(18) $ROD = 5 \times G/S$ for 3°

(19) $ROD = \frac{5 \times G/S \times \text{Glide slope angle}}{3}$ more than 3°

(20) Max Range = $\frac{C \times PRI}{2} = \frac{C}{2 \times PRF}$

(21) $PRI = \frac{106}{PRF}$

(22) minimum Range = $\frac{c \times \text{Pulse width}}{2}$

(23) HT of Cloud ^{Above the A/R} ~~Top~~ ~~Base~~ ~~Base~~ = $(\text{tilt} - \frac{1}{2} Bw) \times R \times 101$
^{Below} ~~Base~~ = $(\text{tilt} + \frac{1}{2} Bw) \times R \times 101$

(24) Diff in frequency = Rate of change $\times$ Time taken

(25) Accuracy of Radio Altimeter
 $= 5 \text{ ft} + 3\%$ of Indicated Height

(25) DME Accuracy

95% of the time

$\pm 1.25 \text{ NM} \pm 1.25\%$ of Indicated height Range

(26)

Mili Sec - 10^{-3} Sec

Micro Sec - 10^{-6} Sec

Nano Sec - 10^{-9} Sec

(27)

$$\text{TOW} = \text{DOW} + \text{FOB} + \text{PAYLOAD}$$

Operating Weight

Useful ~~or~~ ~~Board~~ Load

(28) To calculate the Payload

$$\begin{array}{ccc} \text{RTOW} & \text{MLW} & \text{MZFW} \\ & + \text{Flight fuel} & + \text{FOB} \end{array}$$

Minimum Value

Lowest of three = APS + FOB + Payload

(29)

$$\text{USG} \times 3.78 = \text{Litre}$$

$$\text{KG} \times 2.2 = \text{LBS}$$

$$\text{L} \times 8.6 = \text{KG}$$

$$1.2 \times \text{IMG} = \text{USG}$$

$$\text{IMG} \times 10.54 = \text{LBS}$$

$$1 \text{ KT} = 0.515 \text{ m/sec}$$

(30)

$$(\text{SFC}) \text{ Specific fuel Consumption} = \frac{\text{FUEL FLOW (Kgs/hr)}}{\text{Ground Speed (NM/hr)}}$$

$$(31) \text{ specific Ground Range} = \frac{\text{Ground Speed}}{\text{FUEL FLOW}}$$

$$(32) DCP = \frac{DH}{O+H}$$

$$(33) TCP = \frac{DCP}{G/S(O)}$$

$$(34) DPNR = \frac{EOH}{O+H}$$

$$(35) TPNR = \frac{EH}{O+H}$$

$$(36) \text{ Endurance} = \frac{\text{FOB} - \text{Reserve}}{\text{Fuel Consumption}}$$

$$(37) \text{ Still Air Range} = \text{Endurance} \times \text{TAS}$$

$$(38) \text{ Departure} = D_{\text{long}} \times 60 \times \cos \text{lat}$$

$$(39) \text{ Convergency} = D_{\text{long}} \times \sin \text{mean lat}$$

$$(40) \text{ Scale} = \frac{\text{Chart length}}{\text{Earth Distance}}$$

$$(41) \text{ Conversion Angle} = \frac{1}{2} \times \text{Convergency}$$

$$(42) \frac{D_A}{D_B} = \frac{\cos A}{\cos B}$$

(43) Distance = Speed $\times$ Time

(44) Tail/Head wind component = $V \cos \theta$

(45) X-wind component = $V \sin \theta$

V = wind speed

θ = Angle b/w wind direction and RWY HDG

(46) 1 FT/MIN = 0.00508 m/sec

(47) Ground distance = $\frac{\text{Air Distance} \times G/S}{TAS}$

(48) Still Air Distance = Time $\times$ TAS

(49) Time to station = $\frac{\text{Time to Bearing change} \times 60}{\text{No of Bearing change}}$
 $= \frac{TC}{BC} \times 60$

Note If bearing changes by 45° or doubles than time to station would be equal to time in Bearing change.

(50) ROD = $5 \times G/S$ for 3°

(51) ROD = $\frac{5 \times G/S \times \text{Glide slope angle}}{3}$ (More than 3°)

(52) Change in ROD = $5 \times \text{change in } G/S$ for 3°

(53) change in ROD = $\frac{5 \times \text{change in } G/S \times \text{glide slope angle}}{3}$ (more than 3°)

$$(54) \quad ROD = \text{glideslope gradient } (\%) \times G/S$$

$$(55) \quad \text{glideslope gradient } \% = \frac{\text{Height}}{\text{Range}} \times 100 \quad \star NM \times 6080$$

$$(56) \quad ROD = \frac{\text{Difference in Altitude}}{\text{Time in MIN}}$$

$$(57) \quad X = \frac{D \cdot G_{Sr}}{(G_{So} + G_{Sr})}$$

$$(58) \quad \% = \frac{ROC/ROD(V)}{G/S(H)} \times \frac{6000}{6080}$$

$$(59) \quad \text{Total Deviation } \theta = A + B \sin \theta + C \cos \theta$$

$$(60) \quad \text{Coefficient } A = \frac{\text{Deviation on } (N + S + E + W + NE + SE + NW + SW)}{8}$$

$$(61) \quad \text{Coeff } B = \frac{\text{Deviation on East} - \text{Deviation on West}}{2}$$

$$(62) \quad \text{Coeff } C = \frac{\text{Deviation on North} - \text{Deviation on South}}{2}$$

$$(63) \quad (\text{Aggregate error})^2 = (\text{Propagation error})^2 + (\text{Airborne eq error})^2$$